PREFACE

I've had a half-century-long career as a scientist. It's been joyously fun (most of the time), and has given me a powerful perspective on our world and the universe.

As a child and later as a teenager, I was motivated to become a scientist by reading science fiction by Isaac Asimov, Robert Heinlein, and others, and popular science books by Asimov and the physicist George Gamow. To them I owe so much. I've long wanted to repay that debt by passing their message on to the next generation; by enticing youths and adults alike into the world of science, real science; by explaining to nonscientists how science works, and what great power it brings to us as individuals, to our civilization, and to the human race.

Christopher Nolan's film *Interstellar* is an ideal messenger for that. I had the great luck (and it *was* luck) to be involved with *Interstellar* from its inception. I helped Nolan and others weave real science into the film's fabric.

Much of *Interstellar*'s science is at or just beyond today's frontiers of human understanding. This adds to the film's mystique, and it gives me an opportunity to explain the differences between firm science, educated guesses, and speculation. It lets me describe how scientists